Track 5: AI, ML & Data Science

June 2026

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MR ROKESH TECHNOLOGY — Training & Internship Programs

> ***Skill path:*** *Python → NumPy/Pandas → Stats → ML → DL → NLP → CV → GenAI → Deployment*

Related: Master Syllabus | Duration Framework | Project Progression

Track Overview

Who this is for: Aspiring data analysts, ML engineers, and AI application developers.

Prerequisites: Basic math (algebra, statistics helpful); Python taught from Week 1.

Tools & lab environment: Python 3.11+, Jupyter Lab, NumPy, Pandas, scikit-learn, Matplotlib/Seaborn, Keras/TensorFlow, Streamlit/Flask, Google Colab, Kaggle datasets.

Certification targets (6–9 month): Portfolio projects; optional Google ML or AWS ML specialty alignment

Outcomes by Duration

Duration	Hours	Job Readiness	Key Deliverables
2 Weeks	40–50	ML awareness	Live ML dashboard (Streamlit)
1 Month	80–100	Junior analyst basics	Customer segmentation dashboard
3 Months	250–300	Internship-ready	Resume classifier / Sentiment bot
6 Months	500–600	ML engineer entry-level	End-to-end ML pipeline deployed
9 Months	750–900	Industry-ready	Multi-modal AI + ML microservice

Weekly Rhythm (All Durations)

Mon–Thu notebook labs + coding | Fri analysis/project | Sat review | Sun document findings in README

2-Week Crash Course

Mini Project: Live ML dashboard | **Level:** L1

Days	Topics	Project	Self-Check
D1–2	Python for data, NumPy, Pandas	Data cleaning notebook	Load CSV and handle missing values
D3–4	Visualization (Matplotlib/Seaborn), EDA	Sales analysis report	Choose chart type for distribution vs trend
D5–6	ML basics: regression, classification (scikit-learn)	House price / spam classifier	Explain train/test split
D7–8	Model evaluation, cross-validation, metrics	Churn prediction mini model	Precision vs recall trade-off
D9–10	Streamlit dashboard + deploy	**Live ML dashboard**	Deploy to Streamlit Cloud

1-Month Foundation

Capstone: Customer segmentation dashboard | **Level:** L2

Week	Topics	Project
W1	Python + Pandas + visualization	EDA on retail dataset
W2	Statistics + SQL for analytics	HR analytics with SQL
W3	Supervised ML (regression, classification, trees)	Credit risk predictor
W4	Unsupervised (K-Means) + Streamlit deploy	**Customer segmentation dashboard**

W4 acceptance criteria: - [ ] EDA report with 5+ visualizations - [ ] K-Means model with elbow method documentation - [ ] Streamlit app with interactive filters - [ ] Model metrics displayed in dashboard

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3-Month Job Ready

Capstone: Resume classifier / Sentiment bot | **Level:** L3

Month	Topics	Projects
M1	Python, Pandas, visualization, SQL	EDA + HR analytics
M2	Statistics, supervised/unsupervised ML	Churn + credit risk models
M3	NLP (sentiment), feature engineering, deployment (Flask/Streamlit)	**Resume classifier / Sentiment bot**

M3 Capstone acceptance criteria: - [ ] Text preprocessing pipeline documented - [ ] TF-IDF or embeddings for features - [ ] Model F1-score > baseline; confusion matrix included - [ ] Flask/Streamlit API endpoint for predictions

6-Month Professional

TechKnowledgeHub + Simplilearn aligned

Month	Topics	Projects
M1	Python, NumPy, Pandas, visualization	EDA reports (2)
M2	Statistics, SQL, Excel/Power BI	HR analytics + dashboards
M3	ML: regression, classification, trees, clustering	Churn + credit risk
M4	Feature engineering, NLP, TF-IDF, sentiment	Sentiment analysis bot
M5	Deep learning (Keras), CNNs, transfer learning	Image classifier
M6	GenAI (LangChain, GPT API), MLOps, capstone deploy	**End-to-end ML pipeline with live deployment**

M6 Capstone acceptance criteria (L4): - [ ] Data pipeline from raw data to model - [ ] Dockerized inference API - [ ] LangChain or GPT integration for one feature - [ ] Model versioning documented

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9-Month Advanced

Capstones: Multi-modal AI product + Production ML microservice | **Level:** L5

Additional modules: RNNs/LSTMs, time series forecasting, computer vision (object detection), LLM fine-tuning, RAG pipelines, MLOps (MLflow, Docker, AWS SageMaker), reinforcement learning intro.

Capstone 1 — Multi-modal AI Product: Text + image input, unified API, user-facing app
Capstone 2 — Production ML Microservice: Auto-scaling inference, monitoring, A/B model testing

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Assessment Rubric (Track-Specific)

Component	Weight	Criteria
Weekly Quiz	20%	Stats, ML concepts, Python data ops
Lab Completion	40%	Notebooks run; outputs reproducible
Project	30%	Model quality, EDA depth, deployment
Portfolio	10%	Kaggle-style README; live dashboard links

Recommended Resources

- TechKnowledgeHub 6-Month Curricula
- scikit-learn Documentation
- Kaggle Learn
- Streamlit Documentation

Appendix: Mentor Notes

Common pitfalls: Data leakage in train/test; not scaling features; overfitting without validation; deploying without input validation.

Hardware: 8 GB RAM minimum; GPU optional (Colab free tier for DL labs).

Cross-track: Track 2 — Database for SQL analytics; Track 7 — DevOps for MLOps deployment.